

SpeakToDebug

Debug out loud.

Describe a bug out loud — the agent searches your real code, runs your tests, and speaks the root cause and fix back to you.

AssemblyAI Voice Agent API

Universal-3.5 Pro

Interactive Tool Calling

The problem

- ▶ Bugs strike while your hands are busy — mid-edit, mid-review, second screen
- ▶ Chat assistants can *talk* about code, but they have never seen **your** code
- ▶ Pasting context, copying errors, switching windows — every step breaks flow
- ▶ Voice is how you'd explain it to a colleague. So why not to your tools?

How it works

 Browser mic —wss—▶ **AssemblyAI Voice Agent API**

Universal-3.5 Pro STT · LLM reasoning · TTS · turn-taking

| JSON-Schema **tool calling**

 Local tool layer in your **real repo** — results stream back

 Agent **speaks** the root cause and fix — grounded, never invented

Real tools, real code

- ▶ `search_code` — recursive grep across your actual project
- ▶ `read_file` — opens the suspicious lines, line-ranged
- ▶ `run_tests` — executes your real test suite (npm test)
- ▶ `list_files` — maps the project tree
- ▶ Sandboxed: every path is jailed to the workspace; 15-check self-test

Deep AssemblyAI integration

- ▶ **One WebSocket** handles STT + reasoning + voice output + turn-taking
- ▶ Interactive `tool.call` / `tool.result` loop — the core of the demo
- ▶ Session schema tuned: jitter buffer, min_latency for real-time answers
- ▶ Chinese-first: STT keyword biasing + zh-CN voice output, bilingual captions

Status

- ▶ Working prototype — full voice → tools → voice loop on a real repo
- ▶ 90-second demo video: a live Chinese debugging session
- ▶ Open source, MIT licensed

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